

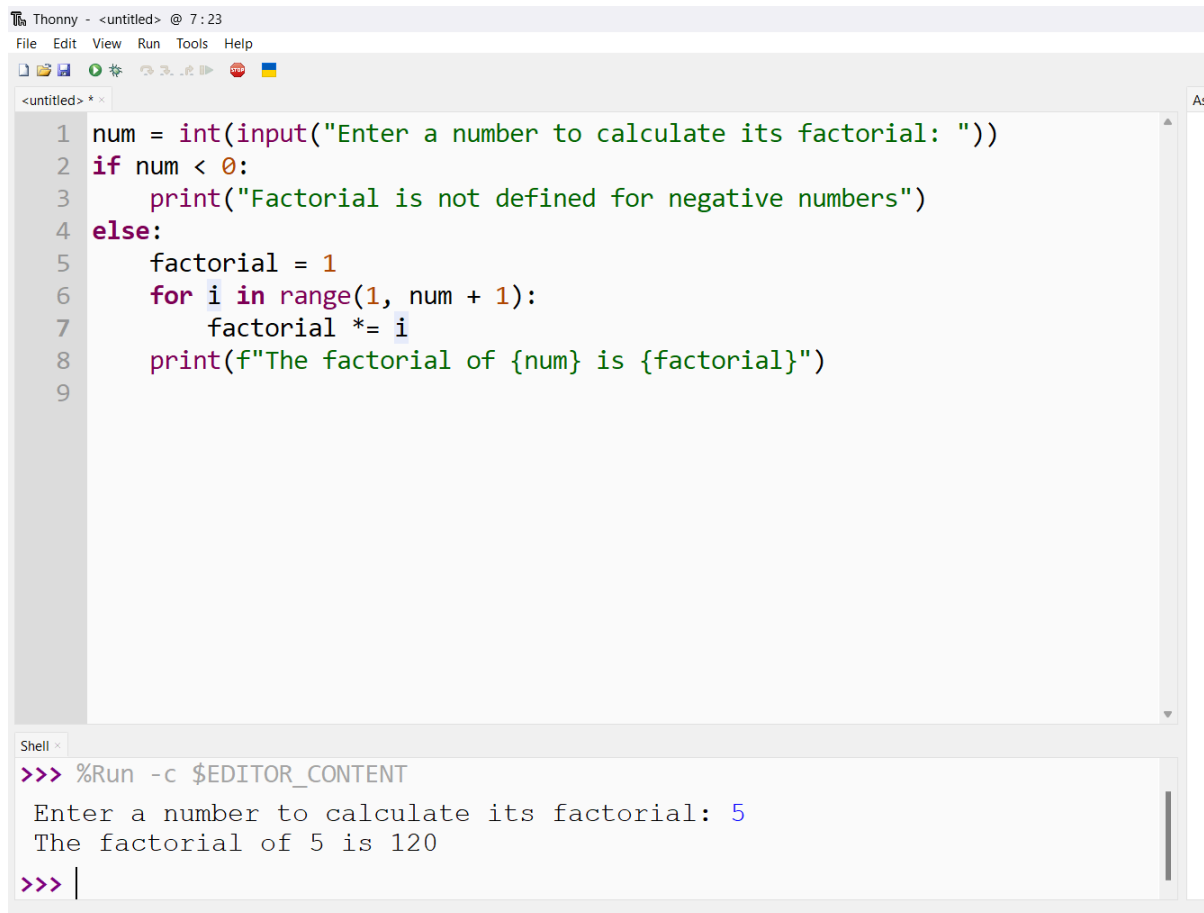
ASSIGNMENT-1

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BATCH-16

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Task 1: AI-Generated Logic Without Modularisation (*Factorial without Functions*)

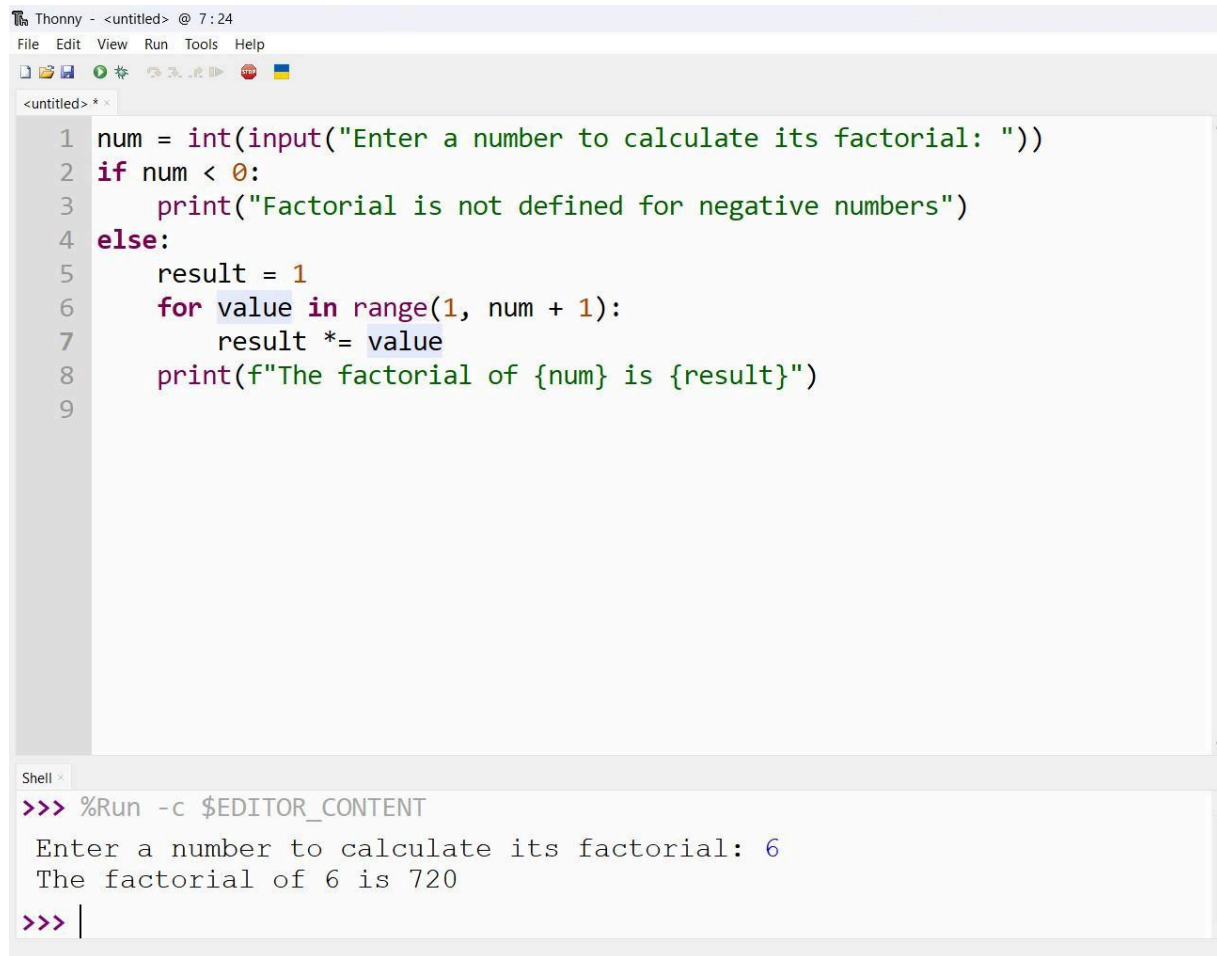


```
Thonny - <untitled> @ 7:23
File Edit View Run Tools Help

<untitled> * <
1 num = int(input("Enter a number to calculate its factorial: "))
2 if num < 0:
3     print("Factorial is not defined for negative numbers")
4 else:
5     factorial = 1
6     for i in range(1, num + 1):
7         factorial *= i
8     print(f"The factorial of {num} is {factorial}")
9

Shell x
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 5
The factorial of 5 is 120
>>> |
```

Task 2: AI Code Optimization & Cleanup (*Improving Efficiency and Readability*)

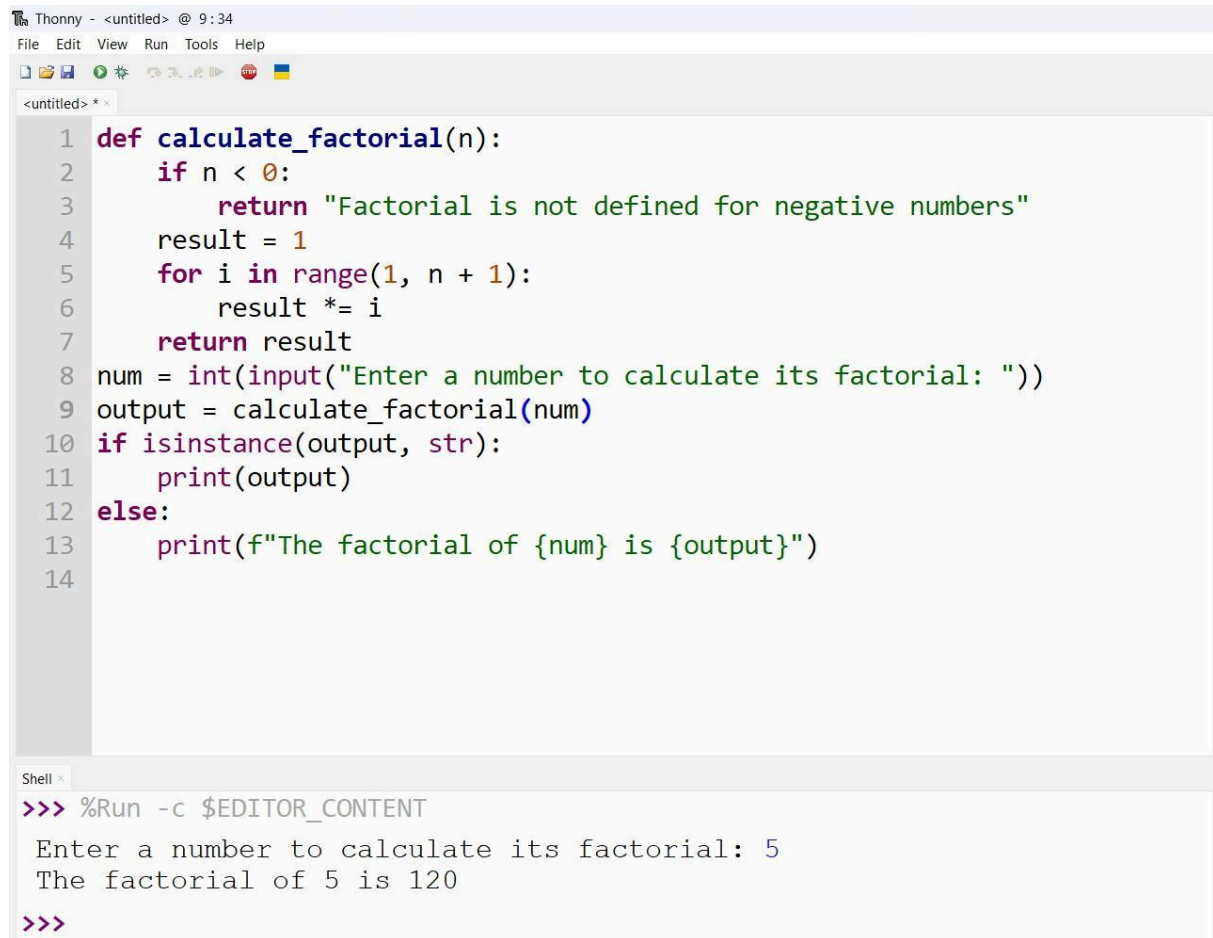


The image shows a screenshot of the Thonny Python IDE. The top window, titled "<untitled> @ 7:24", contains a Python script for calculating factorials. The script uses an if-else structure to handle negative numbers and a for loop to calculate the factorial for positive numbers. The bottom window, titled "Shell", shows the execution of the script, where the user enters the number 6 and the program outputs the factorial result, 720.

```
1 num = int(input("Enter a number to calculate its factorial: "))
2 if num < 0:
3     print("Factorial is not defined for negative numbers")
4 else:
5     result = 1
6     for value in range(1, num + 1):
7         result *= value
8     print(f"The factorial of {num} is {result}")
9
```

```
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 6
The factorial of 6 is 720
>>> |
```

Task 3: Modular Design Using AI Assistance (Factorial with Functions)



The screenshot shows the Thonny Python IDE interface. The top menu bar includes File, Edit, View, Run, Tools, and Help. Below the menu is a toolbar with icons for file operations and execution. The main editor window, titled '<untitled> *', contains the following Python code:

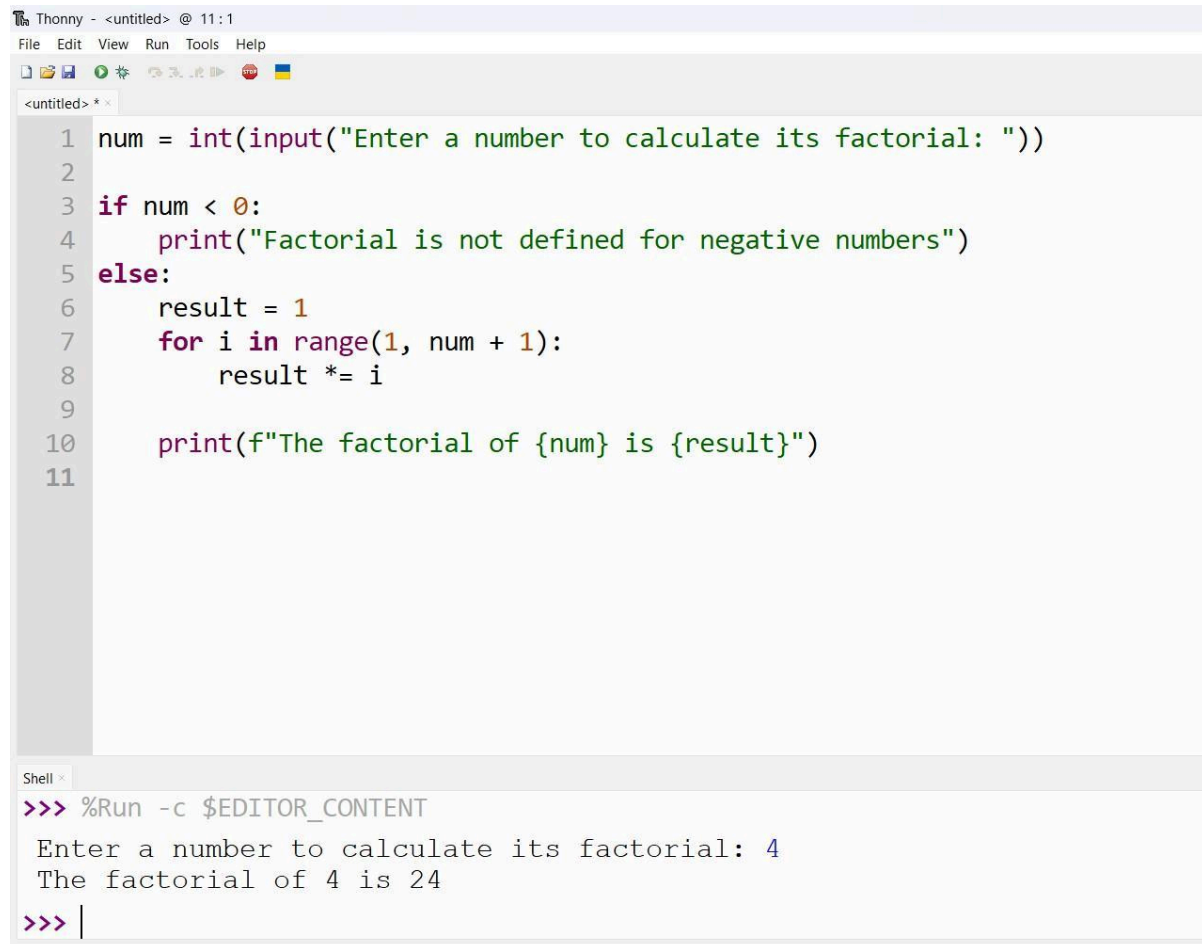
```
1 def calculate_factorial(n):
2     if n < 0:
3         return "Factorial is not defined for negative numbers"
4     result = 1
5     for i in range(1, n + 1):
6         result *= i
7     return result
8 num = int(input("Enter a number to calculate its factorial: "))
9 output = calculate_factorial(num)
10 if isinstance(output, str):
11     print(output)
12 else:
13     print(f"The factorial of {num} is {output}")
14
```

Below the editor is a Shell window titled 'Shell x'. It shows the command prompt with the following interaction:

```
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 5
The factorial of 5 is 120
>>>
```

Task 4: Comparative Analysis – Procedural vs Modular AI Code

Procedural (Without Function):-

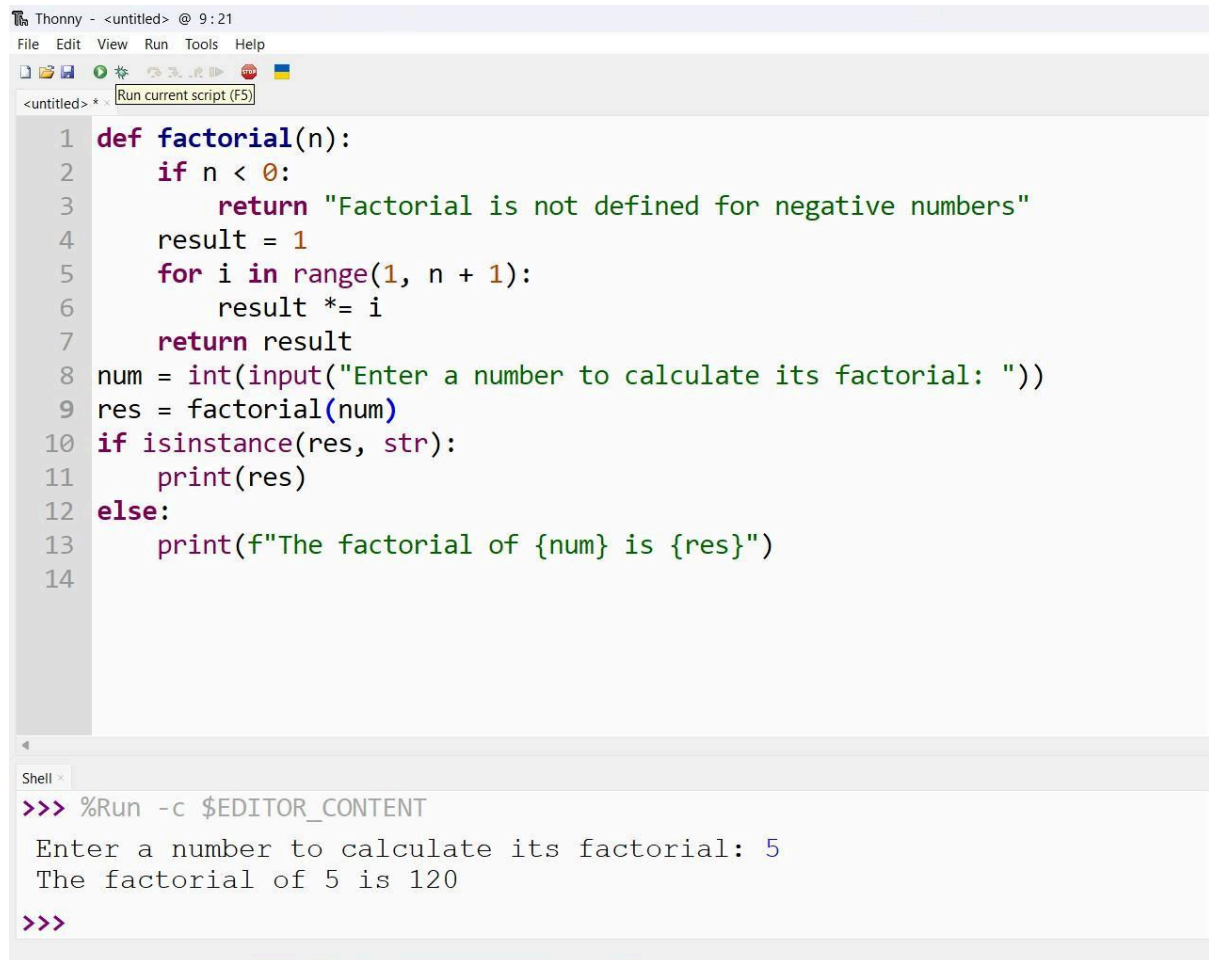


The screenshot shows the Thonny IDE interface. The top bar indicates the file is named '<untitled>' and is at line 11, column 1. The menu bar includes File, Edit, View, Run, Tools, and Help. Below the menu is a toolbar with icons for file operations, running, and debugging. The main editor window shows a Python script for calculating factorials. The script uses `int(input())` to get user input, an `if` statement to check for negative numbers, and a `for` loop to calculate the factorial. The output is printed using an f-string. The bottom panel, titled 'Shell', shows the command `>>> %Run -c $EDITOR_CONTENT` and the program's execution output: 'Enter a number to calculate its factorial: 4' followed by 'The factorial of 4 is 24'. The prompt `>>> |` is visible at the end of the shell output.

```
1 num = int(input("Enter a number to calculate its factorial: "))
2
3 if num < 0:
4     print("Factorial is not defined for negative numbers")
5 else:
6     result = 1
7     for i in range(1, num + 1):
8         result *= i
9
10    print(f"The factorial of {num} is {result}")
11
```

```
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 4
The factorial of 4 is 24
>>> |
```

Modular (With Function)



The screenshot shows the Thonny Python IDE interface. The top menu bar includes File, Edit, View, Run, Tools, and Help. Below the menu is a toolbar with icons for file operations and running code. The main editor window displays a Python script for calculating factorials. The script defines a function `factorial(n)` that returns an error message for negative numbers and calculates the factorial for non-negative numbers using a loop. It then prompts the user for a number, calls the function, and prints the result or an error message. The bottom panel shows a shell window where the script is executed, displaying the input '5' and the output 'The factorial of 5 is 120'.

```
1 def factorial(n):
2     if n < 0:
3         return "Factorial is not defined for negative numbers"
4     result = 1
5     for i in range(1, n + 1):
6         result *= i
7     return result
8 num = int(input("Enter a number to calculate its factorial: "))
9 res = factorial(num)
10 if isinstance(res, str):
11     print(res)
12 else:
13     print(f"The factorial of {num} is {res}")
14
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 5
The factorial of 5 is 120
>>>
```

Task 5: AI-Generated Iterative vs Recursive Thinking

Iterative Approach

```
Thonny - <untitled> @ 8:23
File Edit View Run Tools Help
<untitled> * x
1 # Iterative factorial program
2 num = int(input("Enter a number to calculate its factorial: "))
3 if num < 0:
4     print("Factorial is not defined for negative numbers")
5 else:
6     factorial = 1
7     for i in range(1, num + 1):
8         factorial *= i
9     print(f"The factorial of {num} is {factorial}")
10

Shell x
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 5
The factorial of 5 is 120
>>> |
```

Recursive Approach

Thonny - <untitled> @ 1:30

File Edit View Run Tools Help

<untitled> * x

```
1 # Recursive factorial program
2 def factorial(n):
3     if n <= 1:
4         return 1
5     return n * factorial(n - 1)
6 num = int(input("Enter a number to calculate its factorial: "))
7 if num < 0:
8     print("Factorial is not defined for negative numbers")
9 else:
10    result = factorial(num)
11    print(f"The factorial of {num} is {result}")
12
```

Shell x

```
>>> %Run -c $EDITOR_CONTENT
Enter a number to calculate its factorial: 5
The factorial of 5 is 120
>>> |
```